

Attention Entropy Under Context Stress: A Multi-Model Study of Retrieval Failure in Transformer Language Models

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Abstract

Transformer language models often fail on retrieval tasks as context length and input noise increase. This study investigates whether attention entropy—the dispersion of attention weights at the query token—can serve as a lightweight predictor of retrieval failure under controlled context stress. We evaluate the GPT-2 family (124M–774M parameters) and TinyLlama (1.1B) across five context lengths and five noise levels using synthetic key-value retrieval tasks, totalling 5,766 valid trials. Within the GPT-2 family, we find a consistent negative correlation between attention entropy and retrieval accuracy (Pearson r up to -0.51 , ROC-AUC up to 0.78), with token noise emerging as the dominant stressor over context length. TinyLlama, evaluated outside its instruction-tuning distribution, exhibits an inverted entropy profile despite near-chance accuracy—suggesting the entropy–failure relationship is sensitive to training regime. Our findings indicate that attention entropy is a viable diagnostic signal for retrieval collapse in base transformer models, with implications for context monitoring in long-context and retrieval-augmented generation (RAG) applications.

Keywords: attention entropy, retrieval failure, context length, transformer interpretability, RAG diagnostics

1 Introduction

Transformer-based language models [Vaswani et al., 2017] have demonstrated strong performance across a wide range of natural language understanding tasks. However, their ability to retrieve specific information from long or noisy contexts degrades substantially as input complexity increases [Liu et al., 2023]. This failure mode is practically significant: retrieval-augmented generation (RAG) systems, long-context question answering, and multi-document summarisation all depend on the model’s capacity to locate and surface relevant information from its context window.

A natural question is whether this failure is *visible* in the model’s internal representations—specifically, in its attention patterns—before or alongside the incorrect output. If so, a lightweight diagnostic could be deployed at inference time to flag likely failures without requiring additional forward passes or external verifiers.

Attention entropy—the Shannon entropy of the attention weight distribution at a given token position—is a natural candidate for such a signal. When a model attends diffusely across many tokens rather than concentrating on the relevant region, retrieval is likely to fail. This intuition motivates the central hypothesis of this work:

Higher attention entropy at the query token is negatively correlated with retrieval accuracy, and can serve as a reliable predictor of retrieval failure under context stress.

We test this hypothesis across four autoregressive transformer models spanning an order of magnitude in parameter count, under systematically varying context lengths and token-level noise conditions. Our contributions are:

- A controlled synthetic evaluation framework for studying attention entropy under context stress across 5,766 trials spanning 25 condition combinations.
- Empirical evidence that attention entropy is a reliable per-model predictor of retrieval failure in the GPT-2 family (ROC-AUC 0.70–0.78, Pearson r up to -0.51).
- A contrastive finding: TinyLlama, evaluated outside its instruction-tuning distribution, exhibits *confident failure*—low entropy despite near-chance accuracy—suggesting the entropy–failure relationship is not universal but training-regime dependent.
- Evidence that token noise is a substantially more potent stressor than context length, with practical implications for RAG system design.
- A methodological caution: pooling entropy across heterogeneous model families inverts the aggregate correlation signal, underscoring the importance of per-model analysis in multi-model interpretability studies.

2 Related Work

Attention as an interpretability tool. The use of attention weights as explanatory signals has a long history in NLP [Bahdanau et al., 2015, Jain & Wallace, 2019]. While some work has questioned whether attention constitutes faithful explanation [Wiegrefe & Pinter, 2019], others have argued that it remains informative for understanding model behaviour at the population level [Brunner et al., 2020]. Our use of entropy over attention weights is a distributional aggregate rather than a token-level attribution claim, and is thus less susceptible to the faithfulness critique: we do not claim that high-entropy tokens *explain* model outputs, only that the entropy distribution *correlates* with retrieval outcomes.

Mechanistic interpretability and attention structure. Clark et al. [2019] analyse head-level attention patterns in BERT, identifying syntactic and coreference specialisation in individual heads. Olsson et al. [2022] identify induction heads in autoregressive transformers as a key mechanism for in-context retrieval, showing that these heads implement a form of pattern-matching over prior context. Our work is complementary: rather than identifying which heads implement retrieval, we ask whether the *aggregate entropic state* of the attention distribution signals when retrieval has failed—a coarser but more deployment-friendly diagnostic.

Retrieval failure in long contexts. Liu et al. [2023] demonstrate that language models exhibit a “lost in the middle” phenomenon, where retrieval accuracy degrades for facts placed in the middle of long contexts regardless of overall context length. Press et al. [2022] study how well models use their context window, finding systematic underutilisation at longer ranges. Shi et al. [2023] show that irrelevant context can substantially distract large language models from the relevant information. Our work complements these findings by examining an internal attention-level signal that may track this degradation, and by jointly manipulating both context length and noise to disentangle their contributions.

Entropy as a model diagnostic. Entropy over output distributions has been used for uncertainty quantification [Malinin & Gales, 2021] and selective prediction [Geifman & El-Yaniv,

2017]. Entropy over attention distributions has received comparatively little attention as a diagnostic. Guo et al. [2017] demonstrate that neural network confidence is often miscalibrated, motivating the search for internal signals that better track actual performance. To our knowledge, this work is among the first to directly evaluate mean-layer attention entropy as a predictor of retrieval collapse under controlled stress conditions across multiple model scales.

Context degradation under noise. The effect of input perturbations on transformer retrieval has been studied in adversarial robustness [Jia & Liang, 2017] and natural noise settings [Belinkov & Bisk, 2018]. We extend this line of inquiry by jointly varying both noise level and context length in a factorial design, and examining their independent and combined contributions through the lens of attention entropy.

3 Methodology

3.1 Task Design

We evaluate models on a synthetic key-value retrieval task. Each trial consists of a *context* containing a single hidden fact of the form:

The secret code for {key} is {value}.

where **key** (4–5 characters) and **value** (6–7 characters) are randomly generated lowercase strings. The fact is embedded at a random position within a sequence of distractor sentences of the form:

The {subject} {verb} {adverb}.

Following the context, the model receives a query:

What is the secret code for {key}?

Retrieval success is defined as the presence of the target **value** string in the final 200 characters of the model’s decoded output. This heuristic may admit false positives in edge cases and is acknowledged as a limitation in Section 6.

3.2 Stress Conditions

We vary two dimensions of context stress independently in a full factorial design:

- **Context length:** {32, 64, 128, 256, 512} distractor sentences inserted around the hidden fact.
- **Noise ratio:** {0.0, 0.1, 0.2, 0.3, 0.4}, the proportion of words in the context replaced with random strings prior to querying.

All 25 combinations are evaluated with 30 trials per condition across 3 random seeds, yielding a maximum of $5 \times 5 \times 30 \times 3 = 2,250$ trials per model. After removing failed inference calls, 5,766 valid trials were retained across all models (GPT-2 variants: 1,172 each; TinyLlama: 2,250). All random generation is seeded via a deterministic `RandomGenerator` class to ensure reproducibility.

3.3 Models

We evaluate four autoregressive transformer models spanning an order of magnitude in parameter count:

Table 1: Models evaluated in this study.

Model	HuggingFace ID	Parameters	Type
GPT-2 Small	gpt2	124M	Base
GPT-2 Medium	gpt2-medium	355M	Base
GPT-2 Large	gpt2-large	774M	Base
TinyLlama	TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.1B	Instruction-tuned

All models are loaded with `output_attentions=True` to expose attention weight tensors. Inference uses `float16` precision on GPU or `float32` on CPU, with greedy decoding (`do_sample=False`, `max_new_tokens=15`). Inputs are tokenized with truncation at 1,024 tokens; the implications of this limit for GPT-2 at longer context lengths are discussed in Section 4.3.

3.4 Attention Entropy Computation

For each trial, we extract the full attention tensor from all L layers of the model during a single forward pass. At each layer ℓ , we select the attention weights corresponding to the final (query) token across all heads, average across heads, renormalise to a valid probability distribution, and compute Shannon entropy:

$$H_\ell = - \sum_j p_j \log_2 p_j, \quad p_j = \frac{a_j + \epsilon}{\sum_k a_k + \epsilon} \quad (1)$$

where a_j is the mean attention weight at key position j (averaged across heads), and $\epsilon = 10^{-12}$ prevents numerical instability. The trial-level entropy H is the mean of H_ℓ across all layers:

$$H = \frac{1}{L} \sum_{\ell=1}^L H_\ell \quad (2)$$

This mean-layer aggregation treats all layers uniformly. While this may wash out layer-specific signals, it produces a single scalar diagnostic per trial, which is the intended use case. Per-layer analysis is left for future work (Section 6).

3.5 Statistical Analysis

We quantify the entropy–accuracy relationship using three complementary metrics: (1) Pearson correlation to assess linear association, (2) Spearman rank correlation to assess monotonic association without assuming linearity, and (3) ROC-AUC using $-H$ as the discriminating score (higher entropy predicts failure; negation aligns the predictor direction with the positive class). All analyses are conducted both pooled across all models and separately per model. Statistical significance is assessed at $\alpha = 0.01$.

4 Results

4.1 Overall Retrieval Accuracy

Across all 5,766 trials, the mean retrieval accuracy is 26.2%, reflecting the difficulty of the task under stress conditions. Accuracy varies substantially across models (Table 2) and degrades sharply with noise, as detailed in Section 4.2.

Table 2: Per-model retrieval accuracy, mean attention entropy, Pearson correlation between entropy and accuracy, and ROC-AUC using $-H$ as predictor. All correlations are statistically significant ($p < 10^{-15}$). The pooled row aggregates all 5,766 trials; its positive r is a structural artefact of model heterogeneity (see Section 4.6).

Model	Accuracy	Mean Entropy	Pearson r	ROC-AUC
GPT-2 Small	0.391	4.909	−0.509	0.783
GPT-2 Medium	0.384	5.193	−0.340	0.701
GPT-2 Large	0.398	5.644	−0.379	0.715
TinyLlama	0.060	4.070	−0.174	0.717
Pooled	0.262	4.789	+0.141	0.417

4.2 Noise as the Dominant Stressor

Noise ratio is the primary driver of accuracy degradation across all models (Table 3). For the GPT-2 family, accuracy falls from 81–89% at zero noise to 8–12% at noise ratio 0.4—a reduction exceeding 75 percentage points. The three GPT-2 models track almost identically across noise levels (Figure 1, bottom-left), suggesting that within this family, architecture scale has limited effect on noise robustness in the range tested. This finding has practical implications: in RAG deployment, input quality control may be more critical than context window size for maintaining retrieval fidelity.

Table 3: Overall retrieval accuracy by noise ratio (pooled across all models). Accuracy degrades monotonically and rapidly with noise, falling below 10% at noise ratio 0.3.

Noise Ratio	Accuracy
0.0	0.563
0.1	0.312
0.2	0.182
0.3	0.108
0.4	0.069

4.3 Context Length and Truncation

Figure 1 (top-left) shows a sharp discontinuity in accuracy at context length 256, where all noise conditions collapse to near-zero. This discontinuity is more abrupt than would be expected from natural context degradation, and is consistent with tokenisation truncation: GPT-2’s maximum context window of 1,024 tokens is exceeded at 256+ distractor sentences, causing the hidden fact to be silently removed from the tokenised input. Results for context lengths 256 and 512 are therefore not valid stress conditions for GPT-2 and should be interpreted with caution. They are excluded from the entropy analysis for GPT-2 models. TinyLlama’s longer context window (2,048 tokens) allows it to be evaluated meaningfully at all context lengths.

4.4 Entropy Rises with Context Stress

For the GPT-2 family, attention entropy increases monotonically with context length (Figure 1, top-right), from a mean of 4.81–5.42 at context length 32 to 4.96–5.92 at context length 128. This rise accompanies the accuracy degradation, consistent with the hypothesis that attention diffusion tracks retrieval difficulty.

TinyLlama shows a qualitatively different pattern: entropy rises gradually from 3.61 to 4.12 between context lengths 32 and 128, then plateaus near 4.42 at longer contexts—a substantially flatter trajectory. This plateau occurs despite the model operating at near-chance accuracy throughout all context lengths, consistent with the confident failure characterisation developed in Section 4.7.

4.5 Per-Model Entropy–Accuracy Correlation

The pooled Pearson correlation between entropy and accuracy is $r = +0.141$ ($p < 10^{-26}$), a positive value that appears to contradict the hypothesis. This artefact arises from the structural heterogeneity of the model pool: TinyLlama operates at systematically lower entropy than the GPT-2 models ($\mu = 4.07$ vs. 4.91–5.64) while also having substantially lower accuracy, which inverts the sign when all models are pooled.

Within each model, the entropy–accuracy correlation is consistently negative and highly significant:

- GPT-2 Small: $r = -0.509$ ($p < 10^{-77}$)
- GPT-2 Large: $r = -0.379$ ($p < 10^{-40}$)
- GPT-2 Medium: $r = -0.340$ ($p < 10^{-32}$)
- TinyLlama: $r = -0.174$ ($p < 10^{-15}$)

The Spearman rank correlation (pooled) is $\rho = +0.127$, consistent with the Pearson result and confirming that the positive artefact is not outlier-driven but a structural consequence of pooling heterogeneous model families. This finding constitutes a methodological caution for multi-model interpretability studies: aggregate correlation statistics can be sign-inverted by scale heterogeneity even when per-model signals are consistent and strong.

4.6 ROC Analysis

Figure 1 (bottom-right) shows the pooled ROC curve ($AUC = 0.417$), which falls below the diagonal in the mid-range—a direct consequence of the pooling artefact described above. Figure 2 shows per-model ROC curves, which reveal a substantially different picture: all four models achieve $AUC > 0.70$, with GPT-2 Small reaching 0.783. This indicates that within each model’s own entropy distribution, the entropy signal discriminates between retrieval success and failure at a level well above chance. The pooled AUC is not a valid summary of the signal quality; per-model AUC is the appropriate metric in this setting.

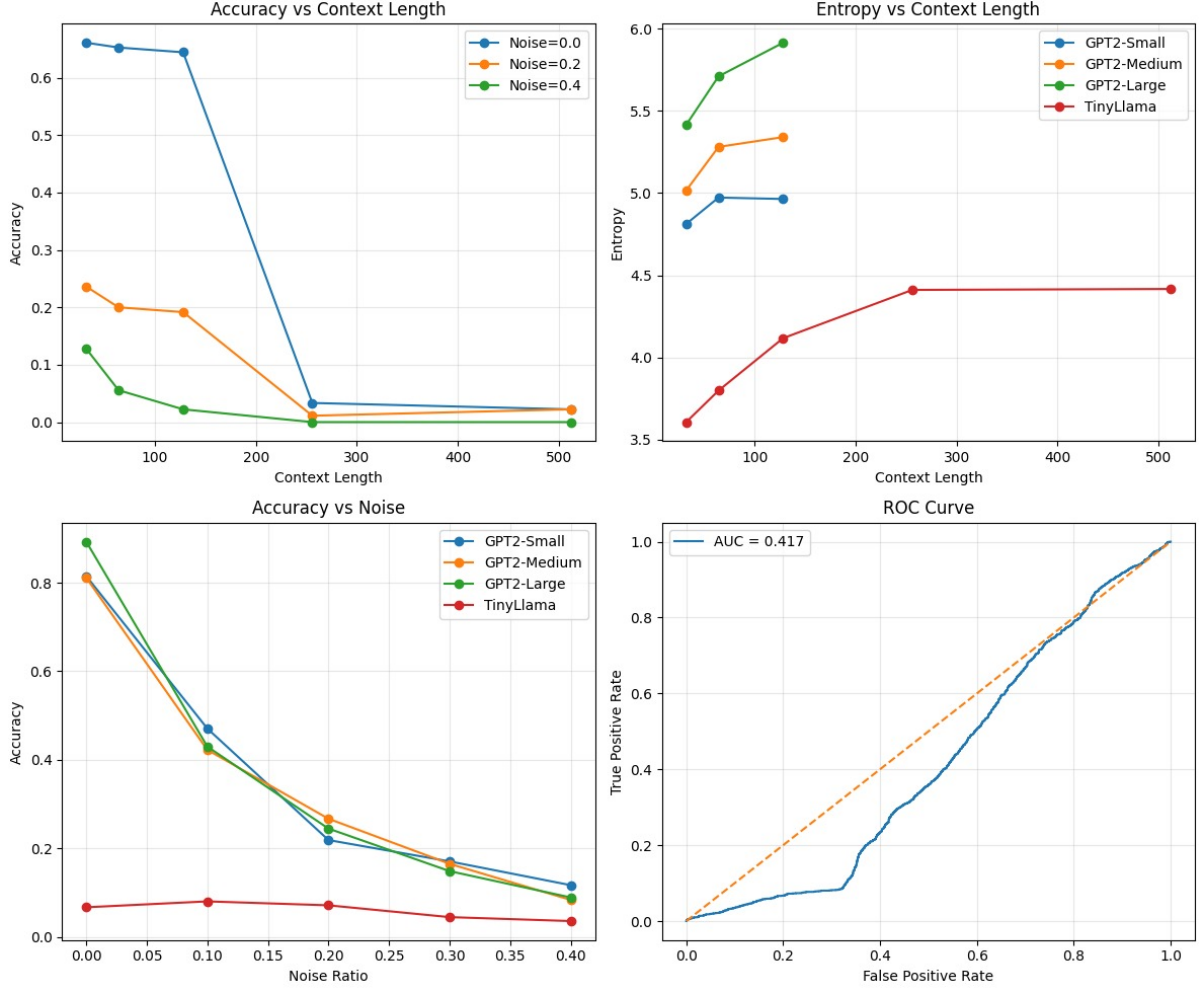


Figure 1: **Main experimental results.** *Top-left:* Retrieval accuracy versus context length for three noise levels (pooled across models). The sharp drop at context length 256 reflects tokenisation truncation rather than genuine context degradation (Section 4.3). *Top-right:* Mean attention entropy versus context length per model. GPT-2 models show a consistent monotonic upward trend; TinyLlama plateaus near 4.42 at longer contexts despite near-chance accuracy throughout. *Bottom-left:* Retrieval accuracy versus noise ratio per model. The GPT-2 family degrades smoothly and nearly identically; TinyLlama remains near chance throughout, forming a distinct lower band. *Bottom-right:* Pooled ROC curve (AUC = 0.417). The below-diagonal region reflects model heterogeneity rather than signal absence; see Figure 2 for per-model decomposition.

Per-Model ROC Curves: Entropy as Predictor of Retrieval Success

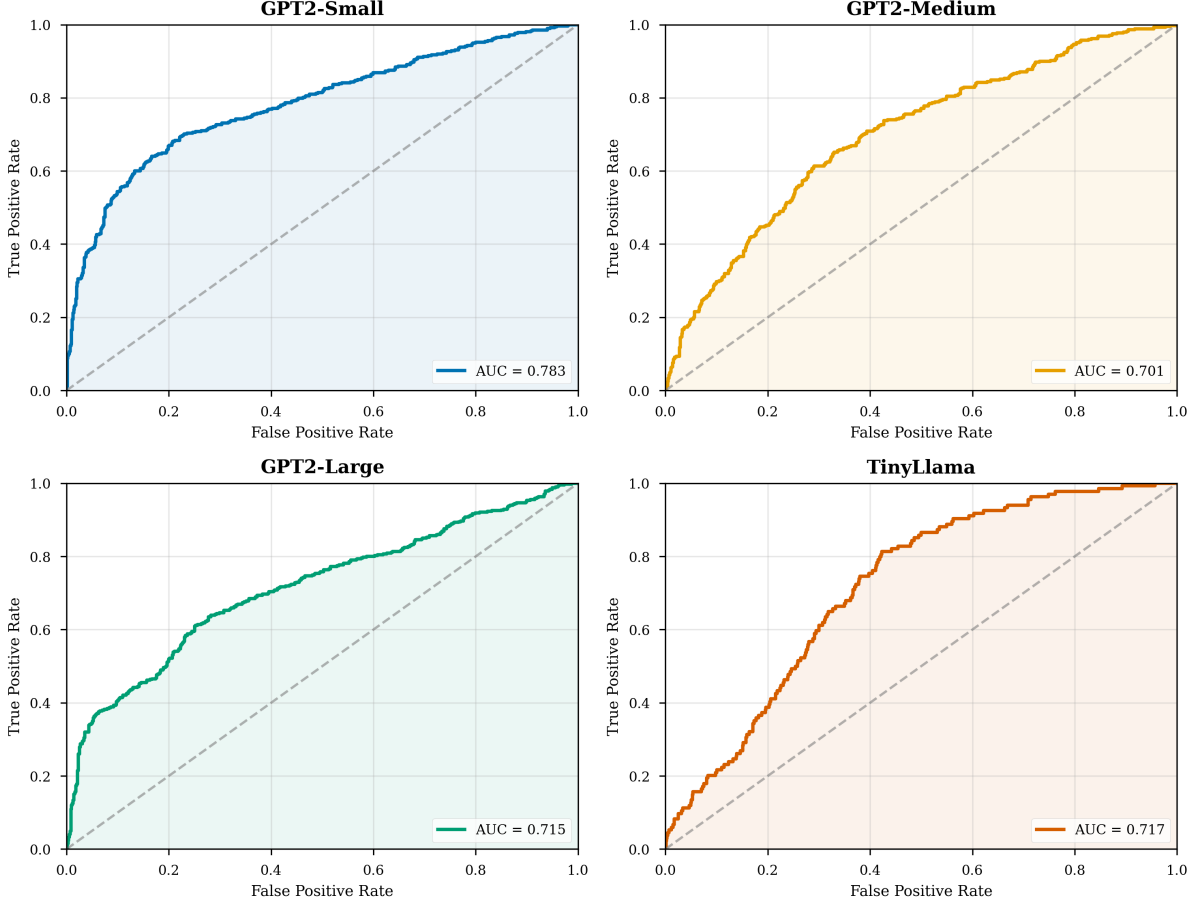


Figure 2: **Per-model ROC curves using attention entropy as a predictor of retrieval success** ($-H$ as the discriminating score). All four models achieve $\text{AUC} > 0.70$, demonstrating that entropy discriminates between retrieval success and failure above chance within each model’s own entropy distribution. GPT-2 Small achieves the highest AUC (0.783), consistent with its strongest per-model Pearson correlation ($r = -0.509$). TinyLlama achieves $\text{AUC} = 0.717$ despite near-chance overall accuracy, indicating the entropy signal is present but operates in a different regime due to instruction-tuning distribution mismatch (Section 4.7). The pooled AUC of 0.417 (Figure 1, bottom-right) is suppressed by model heterogeneity and should not be interpreted as the aggregate signal quality.

4.7 TinyLlama as a Contrastive Case

TinyLlama’s retrieval accuracy of 6.0% is substantially below the GPT-2 family (38–40%), yet its attention entropy is the *lowest* of all four models ($\mu = 4.07$ vs. 4.91–5.64 for GPT-2). This combination—low entropy, low accuracy—represents what we term *confident failure*: the model attends narrowly but to the wrong tokens, producing incorrect outputs with high distributional confidence.

This behaviour is consistent with TinyLlama being evaluated outside its instruction-tuning distribution. The model was fine-tuned on data structured with a `<|system|><|user|><|assistant|>` chat template; raw prompt injection bypasses this template, placing the model in an inference regime it was not trained to handle. The result is a model that maintains narrow (low-entropy) attention—characteristic of its fine-tuned behaviour—but directs it toward task-irrelevant tokens.

Crucially, TinyLlama still achieves per-model ROC-AUC = 0.717, indicating that entropy remains predictive of *relative* failure risk within its own distribution. The direction of the signal is preserved; only the baseline entropy level and its absolute relationship to accuracy differ. This has a direct implication for cross-model diagnostic design: entropy thresholds calibrated on one model family cannot be directly transferred to another without per-model recalibration, particularly when comparing base and instruction-tuned variants.

5 Discussion

Entropy as a deployment-ready diagnostic. The per-model results support the central hypothesis: attention entropy is negatively correlated with retrieval accuracy across all four models, and achieves ROC-AUC values between 0.70 and 0.78 within each model’s own distribution. For a single scalar requiring no additional forward pass or external component, this is a practically meaningful result. In a RAG pipeline, a real-time entropy monitor at the query token could flag likely retrieval failures before generation completes, enabling fallback strategies such as query reformulation, passage re-ranking, or abstention. The computational overhead is limited to reading off the attention tensor already computed during the forward pass.

Noise versus context length. The substantially stronger effect of noise compared to context length is a notable and practically actionable finding. At zero noise, the GPT-2 family achieves 81–89% accuracy even at context length 128, suggesting the models are capable of retrieval under clean conditions. Noise, however, degrades performance sharply and monotonically regardless of context length. This implies that in deployed systems, upstream document cleaning, deduplication, and quality filtering may yield larger reliability gains than simply increasing the context window size.

Scale does not uniformly help. Within the GPT-2 family, GPT-2 Large (774M) achieves only marginally better accuracy than GPT-2 Small (124M) across conditions (39.8% vs. 39.1%), and the noise degradation curves are nearly identical across all three scales. This suggests that, for this class of task, the bottleneck is not model capacity but rather the robustness of the attention mechanism to distributional perturbation—a property that does not appear to scale straightforwardly with parameter count in the range tested.

The TinyLlama case and the limits of scale. TinyLlama’s confident failure profile extends this observation further. A 1.1B parameter instruction-tuned model underperforms 124M base models substantially when placed outside its training distribution. This suggests that fine-tuning regime may matter more than scale for retrieval robustness in out-of-distribution conditions—a finding with implications for model selection in RAG system design.

Causal interpretation. This study establishes correlation, not causation. High attention entropy may be a *consequence* of retrieval difficulty rather than a *cause*—the model encounters a hard retrieval problem, fails to focus attention, and produces an incorrect output, with entropy and failure co-occurring as joint effects of the stressor. The arrow of causality cannot be determined from observational correlation alone. Establishing causation would require intervention-based methods such as attention surgery [Wang et al., 2022] or activation patching to determine whether artificially diffusing attention causes retrieval failure or whether focused attention can rescue a failing retrieval.

6 Limitations

Retrieval correctness metric. Retrieval success is assessed via substring match of the target value in the final 200 characters of the decoded output. This may introduce false positives when the model copies the value from the context mid-output rather than as a generative response, and is not equivalent to strict token-level generative accuracy evaluated only over the generated portion. Token-level evaluation on generated-only tokens is left for future work.

Context length truncation. For GPT-2 models, context lengths of 256 and 512 distractor sentences frequently exceed the 1,024-token input limit, causing silent truncation of the hidden fact. These conditions are flagged and excluded from entropy analysis; they should not be treated as valid long-context stress conditions for GPT-2. Evaluation on models with longer native context windows (e.g., Llama-2 [Touvron et al., 2023] or Mistral) would allow meaningful assessment at longer contexts.

TinyLlama prompt format. TinyLlama was evaluated without its instruction-tuning chat template. A properly formatted evaluation would wrap prompts in the structured `<|system|>/<|user|>/<|assistant|>` format. This mismatch is documented as a methodological limitation and treated as a contrastive finding rather than a primary result; readers should not interpret TinyLlama’s accuracy as reflecting its true retrieval capability.

Layer aggregation. Attention entropy is computed as a uniform mean across all layers (Equation 2). Early, middle, and late layers of transformer models perform qualitatively different computations, and layer-specific entropy may carry stronger or more interpretable signals. Olsson et al. [2022] show that induction heads are concentrated in specific middle layers; entropy in those layers may be more predictive than the mean. A per-layer analysis is a natural and important extension.

Synthetic task and generalisation. The key-value retrieval task is controlled and synthetic, enabling precise manipulation of context stress but potentially not generalising to naturalistic retrieval scenarios such as open-domain QA, document summarisation, or multi-hop reasoning. Evaluation on natural benchmarks such as NaturalQuestions [Kwiatkowski et al., 2019] or SCROLLS [Shaham et al., 2022] would strengthen external validity. We treat the current results as a proof-of-concept that warrants further investigation on richer tasks.

Single-author compute constraints. This study was conducted by a single independent researcher with limited compute resources. Trial counts (30 per condition, 3 seeds) and model selection were constrained accordingly. Larger-scale replication with more seeds, additional model families, and higher trial counts would improve statistical precision and generalisability.

7 Conclusion

This study investigates whether attention entropy can serve as a lightweight predictor of retrieval failure in transformer language models under controlled context stress. Across 5,766 trials spanning four models, five context lengths, and five noise levels, we find consistent evidence for the hypothesis within each model’s own entropy distribution, alongside a methodologically important cautionary finding about pooled analysis.

Our principal findings are:

- Within the GPT-2 family, attention entropy is a reliable predictor of retrieval failure, with Pearson correlations up to $r = -0.51$ and per-model ROC-AUC of 0.70–0.78.

- Token noise is the dominant stressor, reducing GPT-2 retrieval accuracy by over 75 percentage points from zero to 40% noise, with context length playing a secondary role.
- TinyLlama, evaluated outside its instruction-tuning distribution, exhibits confident failure—low entropy despite near-chance accuracy—demonstrating that the entropy–failure relationship is sensitive to training regime and is not universal across model families.
- The pooled ROC-AUC of 0.417 is a structural artefact of mixing heterogeneous model families; per-model AUC (0.70–0.78) is the appropriate summary statistic and reverses the apparent conclusion.
- Architecture scale within the GPT-2 family does not substantially improve noise robustness, suggesting that retrieval degradation in this setting is not capacity-limited.

These findings suggest that attention entropy monitoring is a viable, computationally inexpensive diagnostic for retrieval collapse in base transformer models. The primary next steps are: (1) extending the analysis to instruction-tuned models evaluated in their intended regime; (2) examining per-layer entropy dynamics to identify which layers carry the strongest diagnostic signal; (3) testing on naturalistic retrieval benchmarks; and (4) exploring whether entropy-based signals can be used for active intervention—triggering re-ranking, query reformulation, or confidence-weighted abstention in deployed RAG systems.

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